

BCA-5004 Numerical Methods — Unit I

Roots of Equations: Topic-wise Notes, PYQ Mapping and Corrected Solutions

Course: BCA-5004 Numerical Methods (Semester V)

Source used: 5004-unit-1.pdf. The source PDF is scanned/OCR-extracted; a few mathematical expressions are visibly distorted. This document standardizes the methods, notation, and solutions. Questions whose complete equation is not reliably readable are marked [VERIFY FROM SCAN] rather than guessed.

1. Unit I coverage identified from the question bank

Topic	What to prepare
Roots of equations	Root, bracketing interval, sign change test
Bisection method	Algorithm, convergence proof, numerical iterations, error bound
Regula-Falsi / False Position	Formula, algorithm, numerical solution
Newton-Raphson method	Derivation by Taylor series, algorithm, quadratic convergence, numerical solution
Descartes' rule of signs	Number of possible positive and negative real roots
Elementary equation properties	Equality properties, quadratic formula, zero-product law
Finite differences	First difference; inverse difference question

2. Foundations: roots of an equation

A number α is a root (or zero) of $f(x)=0$ if

$$f(\alpha)=0.$$

Bracketing principle

If f is continuous on $[a, b]$ and

$$f(a) \cdot f(b) < 0,$$

then at least one real root lies in (a, b) . Bisection and Regula-Falsi require such a bracketing interval.

Choosing the method

Situation	Best method
Need guaranteed convergence and have a sign-changing interval	Bisection
Have a sign-changing interval and want faster practical progress	Regula-Falsi
Derivative is easy; a good initial guess is available	Newton-Raphson

Topic A — Bisection Method

Definition

The Bisection Method repeatedly halves an interval $[a, b]$ that contains a root. At every step choose

$$c = (a+b)/2.$$

Then retain the half interval for which the endpoint function values have opposite signs.

Algorithm / steps (exam answer)

1. Write the equation as $f(x)=0$.
2. Find a and b such that $f(a)f(b)<0$.
3. Calculate midpoint $c=(a+b)/2$.
4. Calculate $f(c)$.
5. If $f(c)=0$, then c is the root.
6. If $f(a)f(c)<0$, put $b=c$; otherwise put $a=c$.
7. Repeat Steps 3–6 until the required decimal accuracy is achieved.

Stopping rule

For a starting interval $[a, b]$, after n bisections the interval length is

$$(b-a)/2^n.$$

The midpoint approximation has maximum error at most

$$(b-a)/2^{(n+1)}.$$

To ensure error no more than ϵ , take

$$n \geq \log_2((b-a)/\epsilon) - 1.$$

Why it converges (proof outline)

Let $[a_n, b_n]$ be the interval after the n th step. It always contains a root and

$$b_n - a_n = (b - a) / 2^n \rightarrow 0.$$

The lower endpoints form a bounded non-decreasing sequence and the upper endpoints form a bounded non-increasing sequence. Both approach the same limit α . Since every interval brackets a root and f is continuous, $f(\alpha) = 0$. Hence the method converges.

Corrected solved example

Solve $x^3 - x - 11 = 0$ using bisection.

Take $a=2$, $b=3$:

- $f(2) = 8 - 2 - 11 = -5$
- $f(3) = 27 - 3 - 11 = 13$

Hence a root lies in $(2, 3)$.

Iteration	a	b	$c=(a+b)/2$	$f(c)$	Next interval
1	2.0000	3.0000	2.5000	2.1250	$[2, 2.5]$
2	2.0000	2.5000	2.2500	-1.8594	$[2.25, 2.5]$
3	2.2500	2.5000	2.3750	0.0215	$[2.25, 2.375]$

Continuing produces $x \approx 2.374$ correct to three decimal places.

PYQs: Bisection Method

Year	Corrected/standardized question	What examiner expects
2025–26	Write the steps of Bisection Method.	Seven-step algorithm and sign test
2024–25	Explain Bisection Method. Find a root of $x^3 - x - 11 = 0$.	Definition + iteration table
2023–24	Explain Bisection Method for finding a root.	Procedure and formula
2018	Prove that Bisection Method always converges.	Nested intervals/convergence proof
2021	Find the root of $xe^x - 1 = 0$ between 0 and 1 by Bisection Method.	Bracket and midpoint iterations
2019	Find a real root of an equation by Bisection Method. [Equation unclear in scan]	Standard table
2016	Find the real root of $x^3 - 4x - 9 = 0$ correct to four significant digits by Bisection Method.	Accurate iterations/significant figures

Topic B — Regula-Falsi (False Position) Method

Definition

Regula-Falsi is a bracketing method. Instead of using the midpoint, it joins $(a, f(a))$ and $(b, f(b))$ by a chord and uses the x-intercept of that chord as the next estimate.

Formula

For $f(a)f(b) < 0$, the approximation is

$$x = (a f(b) - b f(a)) / (f(b) - f(a)).$$

Equivalent form:

$$x = a - f(a)(b-a)/(f(b)-f(a)).$$

Algorithm

1. Put the equation in the form $f(x)=0$.
2. Select a, b with $f(a)f(b) < 0$.
3. Compute $x=(a f(b)-b f(a))/(f(b)-f(a))$.
4. Evaluate $f(x)$.
5. If $f(a)f(x) < 0$, replace b by x ; otherwise replace a by x .
6. Repeat until successive estimates give the requested accuracy.

Corrected solved example

Solve $x^2-9x+10=0$ between 2 and 4.

$f(2)=-4$, $f(4)=-10$; therefore these two endpoints do **not** bracket a root. The PDF's printed question/expression appears OCR-corrupted because Regula-Falsi needs opposite signs. Always check $f(a)f(b) < 0$ before applying the method.

A valid illustration is $f(x)=x^2-5$, with $[2, 3]$:

- $f(2)=-1$, $f(3)=4$
- $x_1=(2 \times 4 - 3 \times (-1))/(4 - (-1))=11/5=2.2$
- $f(2.2)=-0.16$, so the next bracket is $[2.2, 3]$.

PYQs: Regula-Falsi

Year	Corrected/standardized question	Topic
2025-26	Explain Regula-Falsi Method. Solve a specified equation between 2 and 4. [Printed polynomial needs scan verification]	Method + numerical
2024-25	Solve the given equation between 2 and 4 by Regula-Falsi Method. [Equation OCR unclear]	Numerical

Year	Corrected/standardized question	Topic
2021	Find $\sqrt{30}$ by Regula-Falsi Method.	Root extraction
2017	Find a real root by False Position Method correct to three decimals. [Equation OCR unclear]	Numerical
2015	Find a real root by Regula-Falsi Method correct to three decimal places. [Equation OCR unclear]	Numerical
Dec. 2013	Find the real root of $3x - \cos x - 1 = 0$ by False Position Method.	Numerical

Topic C — Newton–Raphson Method

Derivation from Taylor series

Suppose x_n is an approximation to a root α of $f(x)=0$. Write

$$\alpha = x_n + h.$$

By Taylor expansion around x_n :

$$f(x_n + h) = f(x_n) + h f'(x_n) + \frac{h^2}{2!} f''(x_n) + \dots$$

Since $f(\alpha) = 0$, and neglecting second and higher powers of the small quantity h ,

$$0 \approx f(x_n) + h f'(x_n).$$

Therefore,

$$h \approx -f(x_n) / f'(x_n)$$

and

$$x_{n+1} = x_n - f(x_n) / f'(x_n).$$

This is Newton–Raphson iteration.

Algorithm

1. Write $f(x)=0$ and calculate $f'(x)$.
2. Choose an initial approximation x_0 near the root.
3. Calculate $x_{n+1} = x_n - f(x_n) / f'(x_n)$.
4. Continue until $|x_{n+1} - x_n|$ is less than the desired tolerance.
5. Do not use an iterate where $f'(x_n) = 0$ or very close to zero.

Convergence / rate

For a simple root α , if $f'(\alpha) \neq 0$, f'' is continuous nearby, and the starting value is sufficiently close, Newton–Raphson converges **quadratically**:

$$e_{n+1} \approx [f''(\alpha) / (2f'(\alpha))] e_n^2, \text{ where } e_n = x_n - \alpha.$$

Meaning: once near the root, the approximate number of correct digits roughly doubles per iteration. It can fail or move away if the initial guess is poor, the derivative is zero/small, or the root is multiple.

Corrected solved example

Find the root of $x^3 - x - 1 = 0$, starting with $x_0 = 1.5$.

$$f(x) = x^3 - x - 1, f'(x) = 3x^2 - 1.$$

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
0	1.5000	0.8750	5.7500	1.3478
1	1.3478	0.1007	4.4499	1.3252
2	1.3252	0.0021	4.2685	1.3247

Hence the root is approximately 1.325 (three decimal places).

PYQs: Newton–Raphson

Year	Corrected/standardized question	What to write
2025–26	Derive Newton–Raphson Method from Taylor series. Use it to solve $x^3 - x - 1 = 0$, starting at $x = 1.5$, correct to three decimals.	Full derivation + table
2024–25	Derive Newton–Raphson equation and discuss convergence.	Formula + quadratic-convergence conditions
2023–24	Write Newton–Raphson algorithm. Find the real root of $3x - \cos x - 1 = 0$ correct to four decimals.	Algorithm + numerical
2021	Discuss/obtain rate of convergence of Newton–Raphson Method.	Quadratic proof/result
2021	Evaluate $\sqrt{12}$ to five decimal places by Newton's iterative method.	Take $f(x) = x^2 - 12$
2019	Find a root of $x \log_{10}(x) - 1.2 = 0$ by Newton–Raphson Method, correct to three decimals.	$f'(x) = \log_{10}(x) + 0.434294 \dots$
2018	Find x for $x^3 - 6x + 4 = 0$ by Newton–Raphson Method.	Choose a close initial root
2017	Find the root of $x^3 - 5x^2 + 2 = 0$ by Newton–Raphson Method.	Numerical
2016	Find the root of $x^2 - 3x + 2 = 0$ by Newton–Raphson Method.	Numerical
2015	Find the real root of $x^3 - 9 = 0$ correct to three decimal places.	Numerical
Dec. 2013	Evaluate the root of $e^x - 3x = 0$ lying between 0 and 1 by Newton–Raphson Method.	Numerical

Topic D — Descartes' Rule of Signs

Statement

For a polynomial $f(x)$ with real coefficients:

- The number of positive real roots is equal to the number of sign changes in $f(x)$, or less than it by an even integer.
- The number of negative real roots is obtained by applying the same rule to $f(-x)$.

Procedure

1. Arrange the polynomial in descending powers of x .
2. Count sign changes in $f(x)$ for possible positive roots.
3. Substitute $-x$ for x to get $f(-x)$.
4. Count sign changes in $f(-x)$ for possible negative roots.

Example

For $f(x) = x^3 - x^2 - x - 1$, signs are +, -, -, -; there is one sign change. Thus it has exactly one positive real root.

For negative roots:

$f(-x) = -x^3 - x^2 + x - 1$, signs are -, -, +, -; there are two sign changes. Hence negative real roots may be 2 or 0.

PYQs

Year	Question
2016, 2017, 2018	Define/Explain Descartes' Rule of Signs.

Topic E — Properties of Equations

Core properties for short answers

- **Addition property:** If $A=B$, then $A+C=B+C$.
- **Multiplication property:** If $A=B$, then $AC=BC$.
- **Division property:** If $A=B$ and $C \neq 0$, then $A/C=B/C$.
- **Zero-product law:** If $XY=0$, then $X=0$ or $Y=0$.
- **Quadratic formula:** For $ax^2+bx+c=0$, $a \neq 0$,
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- **Radical/rational-exponent equations:** Raising both sides to a power can introduce extraneous roots; always verify final answers in the original equation.

PYQ

Year	Question
2016	What are the properties of equations?

Topic F — Finite Difference

First difference

For interval h , the first forward difference is

$$\Delta f(x) = f(x+h) - f(x).$$

For $f(x) = e^x$:

$$\Delta e^x = e^{(x+h)} - e^x = e^x (e^h - 1).$$

Thus, if the first difference is specified as e^x , then the inverse-difference problem must be handled using the corresponding finite-difference operator and interval. The question in the scan is incomplete/OCR-damaged; verify its exact wording before writing a final numerical answer.

PYQ

Question	Status
Find the function whose first difference is e^x (or related expression).	[VERIFY exact expression and interval h from scan]

3. Repeated-question priority

Highest priority

1. Bisection Method: steps, numerical problem, and convergence proof.
2. Newton–Raphson: Taylor-series derivation, algorithm, rate of convergence, and numerical problem.
3. Regula-Falsi: formula and numerical problem.
4. Descartes' Rule of Signs.

Exam writing checklist for numerical questions

- First write $f(x)$ and, for Newton–Raphson, also write $f'(x)$.
- For Bisection/Regula-Falsi, explicitly show $f(a)f(b) < 0$.
- Make a labelled iteration table.
- Round only in the final answer as far as possible.
- State the requested accuracy: e.g., “Therefore, the required root correct to three decimal places is ...”.

4. One-page formula revision

Method	Formula
Bisection	$x_n = (a_n + b_n) / 2$
Bisection error	$ \alpha - x_n \leq (b - a) / 2^{(n+1)}$
Regula-Falsi	$x = (a f(b) - b f(a)) / (f(b) - f(a))$
Newton–Raphson	$x_{(n+1)} = x_n - f(x_n) / f'(x_n)$
Newton local error	$e_{(n+1)} \approx [f''(\alpha) / (2f'(\alpha))] e_n^2$
Sign-change test	$f(a)f(b) < 0$
Quadratic formula	$x = (-b \pm \sqrt{b^2 - 4ac}) / (2a)$

5. Important scan-correction note

The source PDF is a low-quality scanned question bank. In several places, superscripts, minus signs, trigonometric notation and equations have been merged or dropped by OCR. Method names, years and the majority of question topics are readable; exact expressions labelled [VERIFY] should be checked against the page image before submission/exam preparation. No equation has been invented where its source text is unreliable.